



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Juan J. Florez
July 5th, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Methodology:** SpaceX launch records were collected via the SpaceX REST API and Wikipedia web scraping, wrangled into a labeled dataset, explored with visualizations and SQL, mapped interactively with Folium, visualized in a Plotly Dash dashboard, and used to train four tuned classification models (Logistic Regression, SVM, Decision Tree, KNN).
- **Results:** Launch success rose from 0% (2010-2013) to 84% by 2020; KSC LC-39A has the highest site success rate (76.9%); heavier payloads land more reliably in LEO/ISS/Polar orbits; a tuned Decision Tree classifier achieved the best cross-validated accuracy (87.5%) among the four models tested.

Introduction

- **Background:** SpaceX advertises Falcon 9 launches at \$62M, versus \$165M+ from other providers, largely because SpaceX reuses the rocket's first stage. Predicting whether a first stage will land lets us estimate the cost of a launch.
- **Problem:** Can flight and payload characteristics (launch site, payload mass, orbit, booster version, prior flights/reuse) predict whether a Falcon 9 first stage will land successfully, so a competitor could estimate SpaceX's cost when bidding against them?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Combined SpaceX API + Wikipedia scrape into one launch record set
- Perform data wrangling
 - Cleaned outcomes into a binary landing-success label
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Built, tune, and evaluate classification models such as: Logistic Regression, Support Vector Machine, Decision Tree and KNN.

Data Collection

- **Two data sources:** the SpaceX REST API (structured launch, rocket, payload, launchpad, and core records) and a Wikipedia table of historical Falcon 9/Heavy launches (via web scraping).
- **Process:** query each source, normalize nested JSON / HTML tables into flat records, then merge on flight number and date to build a single labeled launch dataset used throughout the rest of the project.

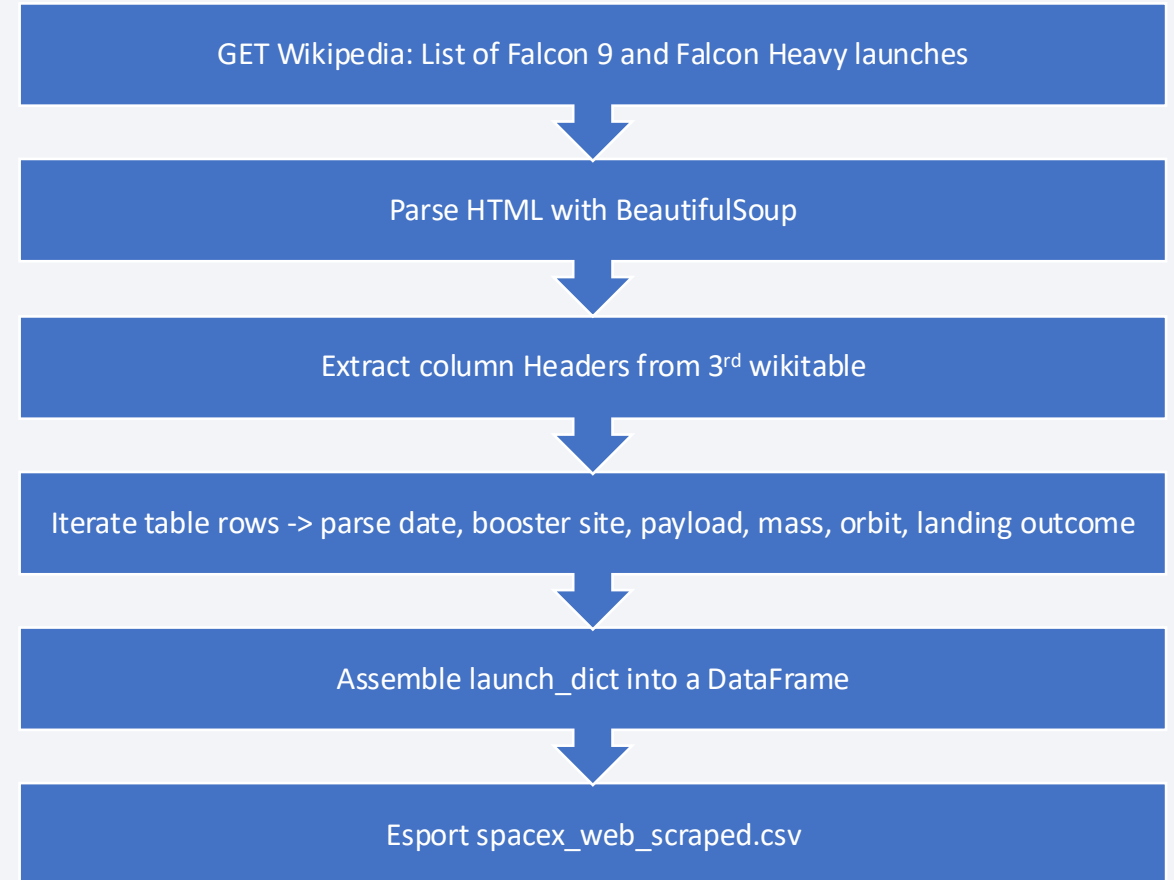
Data Collection – SpaceX API

- **Process:** Requested `api.spacexdata.com/v4/launches/past`, then resolved rocket, payload, launchpad, and core IDs via follow-up API calls to build booster, payload, orbit, site, and landing columns.
- Filtered to Falcon 9 only, re-indexed flight numbers -> `dataset_part_1.csv`.
- **GitHub:**
<https://github.com/jjfloreszw/ibmds/blob/main/01-introduction/API-data-collection.ipynb>



Data Collection - Scraping

- **Process:** Scraped the Wikipedia 'List of Falcon 9 and Falcon Heavy launches' page (9 June 2021 snapshot) with BeautifulSoup, parsed the table, and cleaned reference-link/formatting noise.
- Produced `spacex_web_scraped.csv` as an independent cross-check on the API dataset.
- **GitHub:**
<https://github.com/jjfloreszw/ibmds/blob/main/O1-introduction/web scraping-data-collection.ipynb>



Data Wrangling

- **Processing:** Counted launches per site (CCAFS SLC-40: 55, KSC LC-39A: 22, VAFB SLC-4E: 13) and per orbit (GTO 27, ISS 21, VLEO 14, PO 9, LEO 7, others fewer).
- **Labeling:** Outcome strings (e.g. 'True ASDS', 'False Ocean', 'None None') were mapped to a binary Class label: 1 if the first stage landed successfully, 0 otherwise. Overall success rate = 66.7%.
- **GitHub:** <https://github.com/jjfloreszw/ibmds/blob/main/01-introduction/spacex-data-wrangling.ipynb>

EDA with Data Visualization

- **Charts used:** Scatter plots (Flight Number / Payload Mass vs. Launch Site and vs. Orbit) reveal how success correlates with launch cadence, site, and orbit; a bar chart shows success rate by orbit; a line chart tracks yearly success trend.
- **Why:** Scatter plots expose site- and orbit-specific patterns invisible in aggregate statistics; the bar and line charts summarize trends for quick comparison across categories and time.
- **GitHub:** <https://github.com/jjfloreszw/ibmds/blob/main/O2-EDA/EDA-with-data-viz.ipynb>

EDA with SQL

- Loaded the SpaceX dataset into a SQLite database and used SQL to answer 10 questions:
- Unique launch sites; sample CCA-prefixed records; total NASA (CRS) payload mass; average F9 v1.1 payload mass;
- Date of first successful ground-pad landing; boosters landing on drone ships with 4000-6000 kg payloads;
- Total successful vs. failed mission outcomes; boosters carrying maximum payload; 2015 drone-ship failures; and a ranked count of landing outcomes (2010-06-04 to 2017-03-20).
- **GitHub:** <https://github.com/jjfloreszw/ibmds/blob/main/02-EDA/EDA-with-SQL.ipynb>

Build an Interactive Map with Folium

- **Map objects:** folium.Circle + DivIcon markers for each of the 4 launch sites; a MarkerCluster of green/red markers for every individual launch (success/failure); MousePosition for coordinate lookup; and PolyLines with distance labels from a launch site to the nearest coastline, highway, railway, and city.
- **Why:** Circles/markers localize each site on a world map; color-coded launch markers reveal which sites cluster successes; distance lines test whether proximity to infrastructure relates to site placement.
- **GitHub:** <https://github.com/jjfloreszw/ibmds/blob/main/O3-interactive-visual-analytics-dashboard/interactive-visual-analytics-with-folium.ipynb>

Build a Dashboard with Plotly Dash

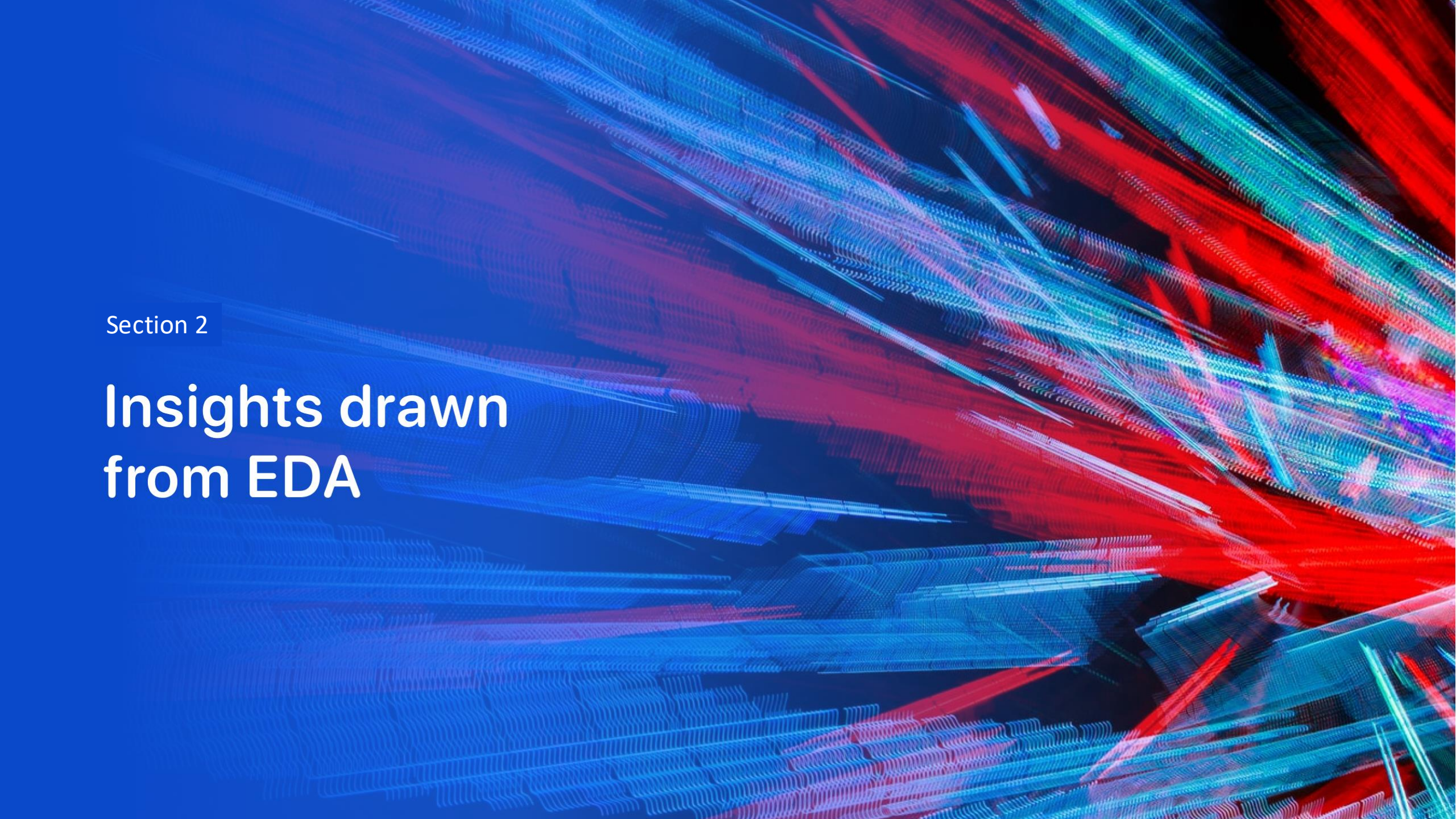
- **Plots & interactions:** A pie chart of successful launch counts by site (with a dropdown to select 'All Sites' or a single site), and a scatter plot of Payload Mass vs. launch outcome colored by Booster Version, filterable with a payload-mass range slider.
- **Why:** The pie chart quickly shows which site contributes most successes; the scatter + slider lets a user isolate which payload ranges and booster versions are most reliable.
- **GitHub:** <https://github.com/jjflorezw/ibmds/blob/main/O3-interactive-visual-analytics-dashboard/spacex-dash-app.py>

Predictive Analysis (Classification)

- **Process:** Standardized features, split into train/test sets, then used GridSearchCV (cv=10) to tune four classifiers: Logistic Regression, SVM, Decision Tree, and KNN.
- **Evaluation:** Compared each model's cross-validated accuracy and test-set accuracy, and inspected confusion matrices to see landing vs. non-landing prediction errors.
- **Best model:** Decision Tree (criterion=gini, max_depth=4, max_features=sqrt) achieved the highest cross-validated accuracy at 87.5%.
- **GitHub:** <https://github.com/jjfloreszw/ibmds/blob/main/O4-predictive-analysis/machine-learning-prediction.ipynb>

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

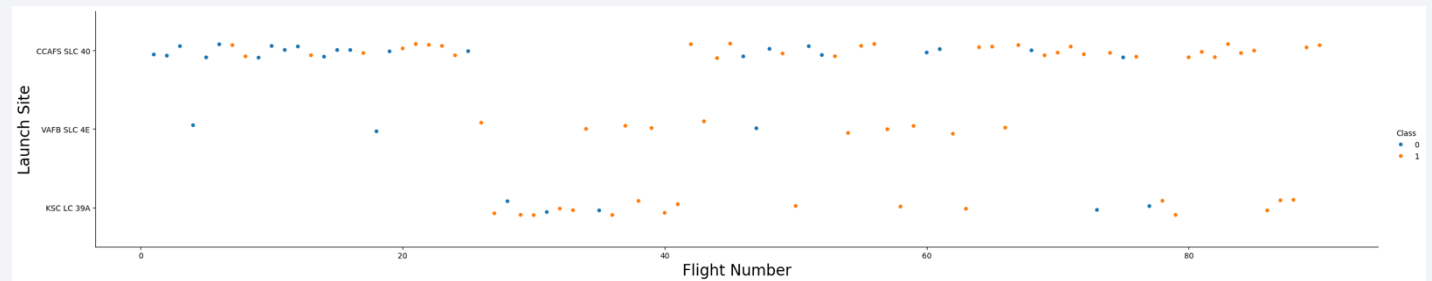
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

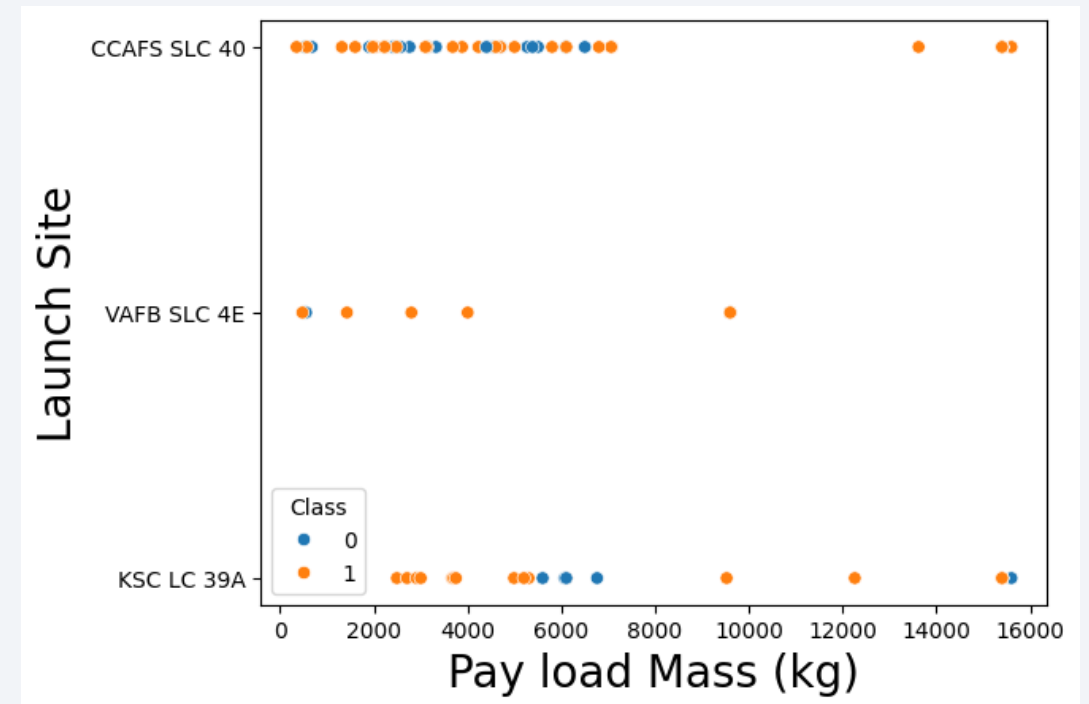
Flight Number vs. Launch Site

- **Explanation:** Early flights launched only from CCAFS SLC-40. As flight number grows, successful landings (blue=0/orange=1 by legend) become common at all three sites, showing SpaceX's learning curve over time.



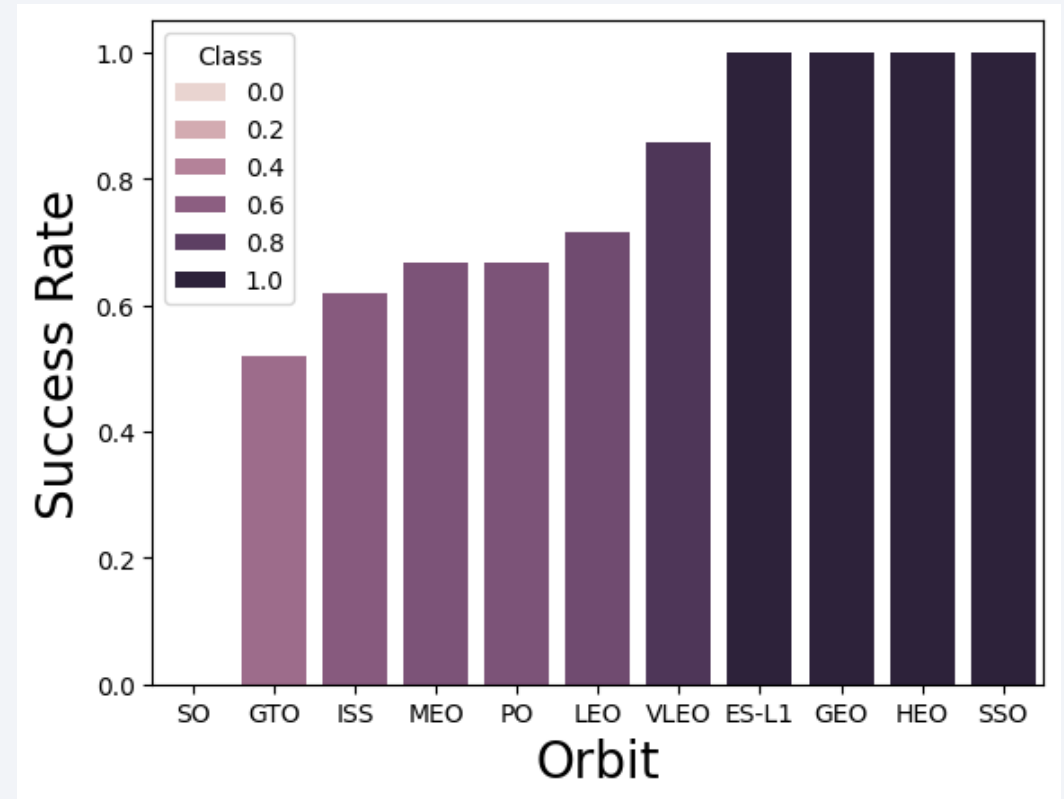
Payload vs. Launch Site

- **Explanation:** VAFB SLC-4E never launches payloads above ~10,000 kg; CCAFS SLC-40 and KSC LC-39A handle the heaviest payloads, and heavy payloads at KSC LC-39A still land successfully.



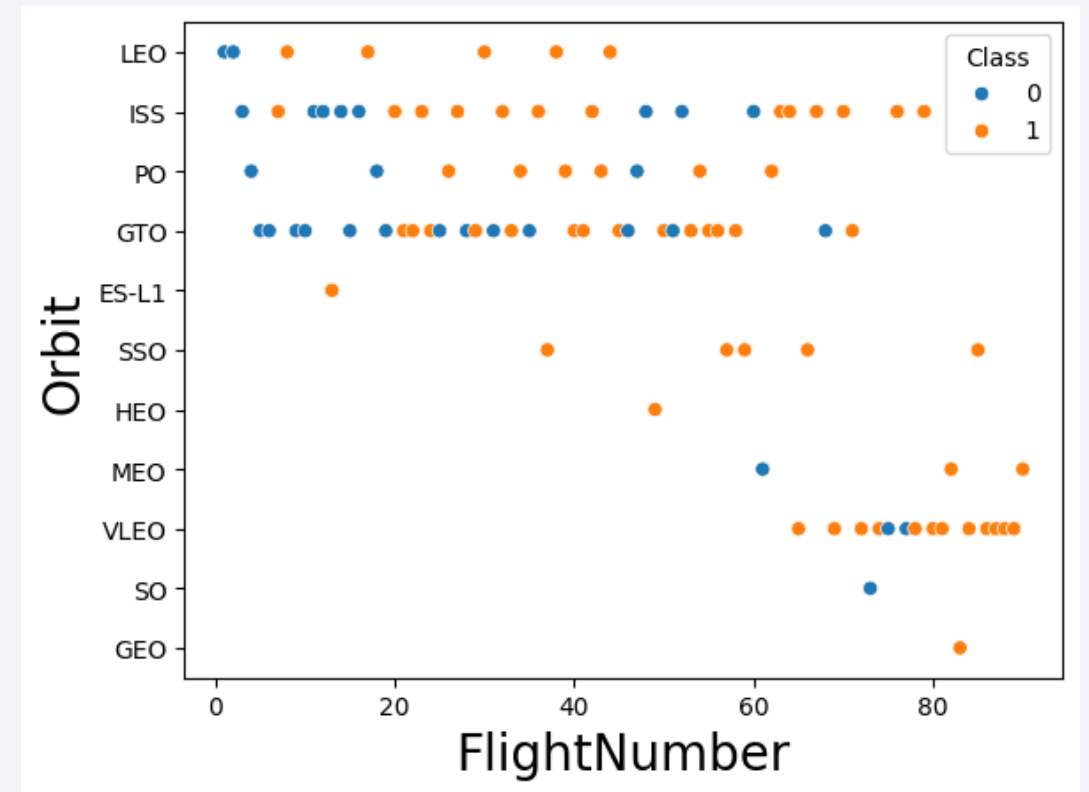
Success Rate vs. Orbit Type

- **Explanation:** ES-L1, GEO, HEO, and SSO orbits show 100% success in this dataset; GTO has the lowest success rate (~52%), while LEO/ISS/VLEO/MEO/PO fall in the 62-72% range.



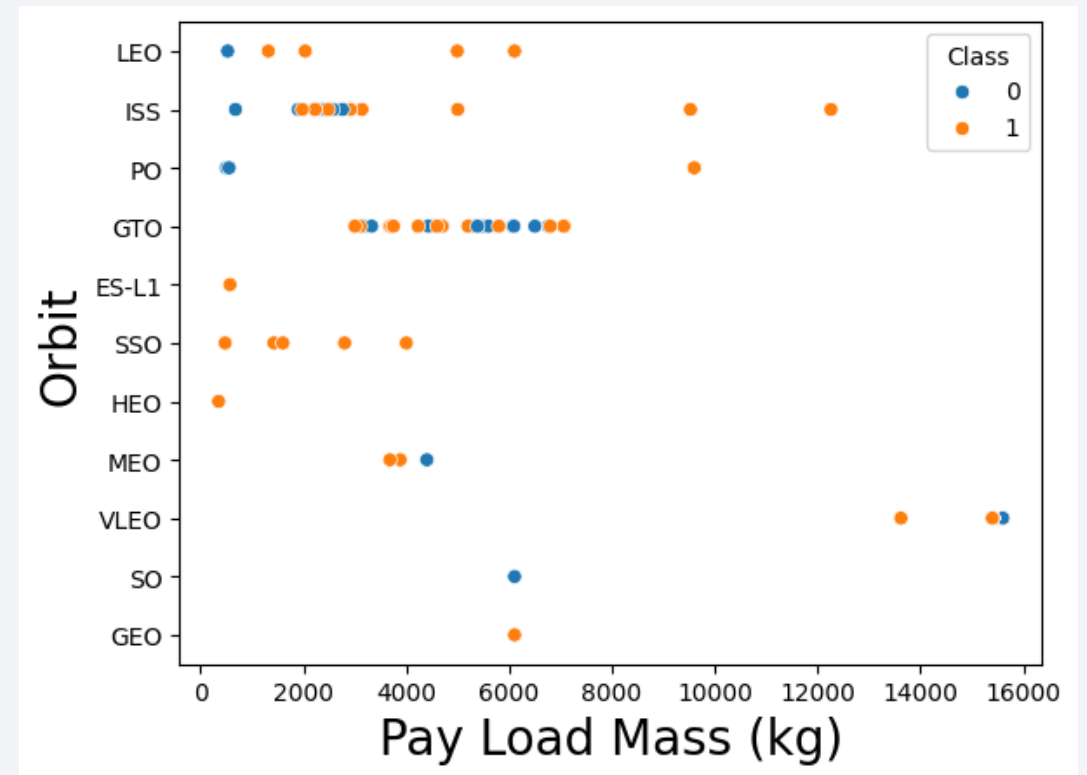
Flight Number vs. Orbit Type

- **Explanation:** In LEO, later flights (higher flight number) succeed almost exclusively, showing a clear learning trend. In GTO, successes and failures are mixed across all flight numbers with no clear trend.



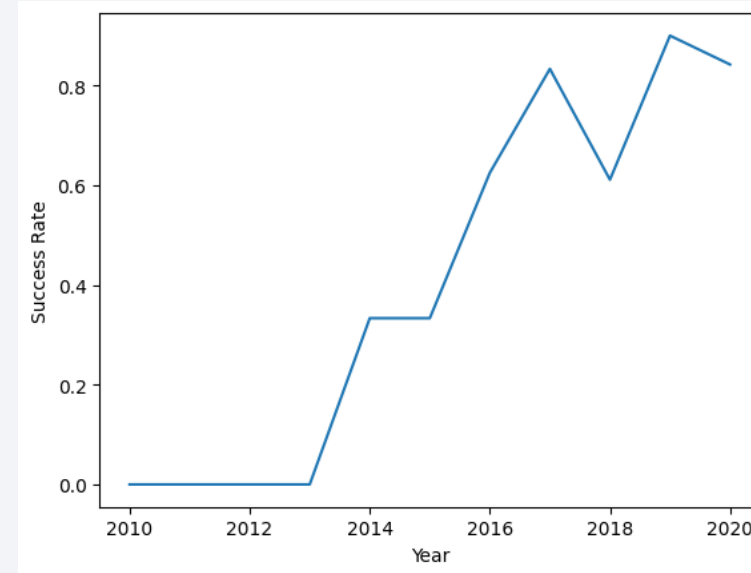
Payload vs. Orbit Type

- **Explanation:** Heavy payloads land successfully in Polar, LEO, and ISS orbits. GTO shows both successes and failures across the payload range, so payload mass alone does not explain GTO outcomes.



Launch Success Yearly Trend

- **Explanation:** Success rate was 0% from 2010-2013, rose sharply after 2013, dipped slightly in 2018, and reached its highest levels (84-90%) in 2019-2020 as SpaceX refined booster recovery.



All Launch Site Names

- **Query:** `SELECT DISTINCT Launch_Site FROM SPACEXTBL`
- **Result:** CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
- **Explanation:** The dataset spans 4 distinct pad/complex names across 3 physical sites (2 CCAFS pads plus KSC and VAFB).

Launch Site Names Begin with 'CCA'

- **Query:** `SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5`
- **Result:** Returned the first 5 CCAFS LC-40 launches (2010-2013), including the Dragon Qualification Unit and early CRS/demo missions.
- **Explanation:** Confirms CCAFS was SpaceX's original and most active launch complex.

Total Payload Mass

- **Query:** `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)'`
- **Result:** 45,596 kg total payload mass carried for NASA's CRS program.
- **Explanation:** NASA CRS resupply missions represent a substantial share of SpaceX's early cargo manifest.

Average Payload Mass by F9 v1.1

- **Query:** `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1'`
- **Result:** 2,928.4 kg average payload mass for the F9 v1.1 booster version.
- **Explanation:** F9 v1.1 was an early, lower-capacity booster generation relative to later Block/FT versions.

First Successful Ground Landing Date

- **Query:** `SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)'`
- **Result:** 2015-12-22 - the first successful ground-pad landing.
- **Explanation:** Marks the milestone that proved first-stage recovery on land was viable, a turning point for reusability.

Successful Drone Ship Landing with Payload between 4000 and 6000

- **Query:** `SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000`
- **Result:** F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
- **Explanation:** All four are Full Thrust (FT) boosters, showing this generation reliably handled mid-range payloads on drone ships.

Total Number of Successful and Failure Mission Outcomes

- **Query:** `SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY 1`
- **Result:** Success: 98 (+ 1 with trailing space), Success (payload status unclear): 1, Failure (in flight): 1.
- **Explanation:** Mission-level (payload delivery) success is over 98% - first-stage landing failures are a separate, much more common outcome.

Boosters Carried Maximum Payload

- **Query:** `SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)`
- **Result:** 12 Block 5 boosters (e.g. B1048.4/.5, B1049.4/.5/.7, B1051.3/.4/.6, B1056.4, B1058.3, B1060.2/.3) tied at the maximum payload mass.
- **Explanation:** The heaviest payloads were consistently flown on the newest Block 5 booster generation.

2015 Launch Records

- **Query:** `SELECT substr(Date,6,2), Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Landing_Outcome='Failure (drone ship)' AND substr(Date,0,5)='2015'`
- **Result:** Jan 2015: F9 v1.1 B1012, CCAFS LC-40; Apr 2015: F9 v1.1 B1015, CCAFS LC-40.
- **Explanation:** Both 2015 drone-ship failures were early F9 v1.1 boosters, before landing technology matured.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **Query:** `SELECT Landing_Outcome, COUNT(*) FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY 1 ORDER BY 2 DESC`
- **Result:** No attempt: 10, Success/Failure (drone ship): 5 each, Success (ground pad): 3, Controlled (ocean): 3, Uncontrolled (ocean): 2, Failure (parachute): 2, Precluded (drone ship): 1.
- **Explanation:** In SpaceX's early years, half of all landing attempts were 'no attempt' - recovery was still experimental before becoming routine.

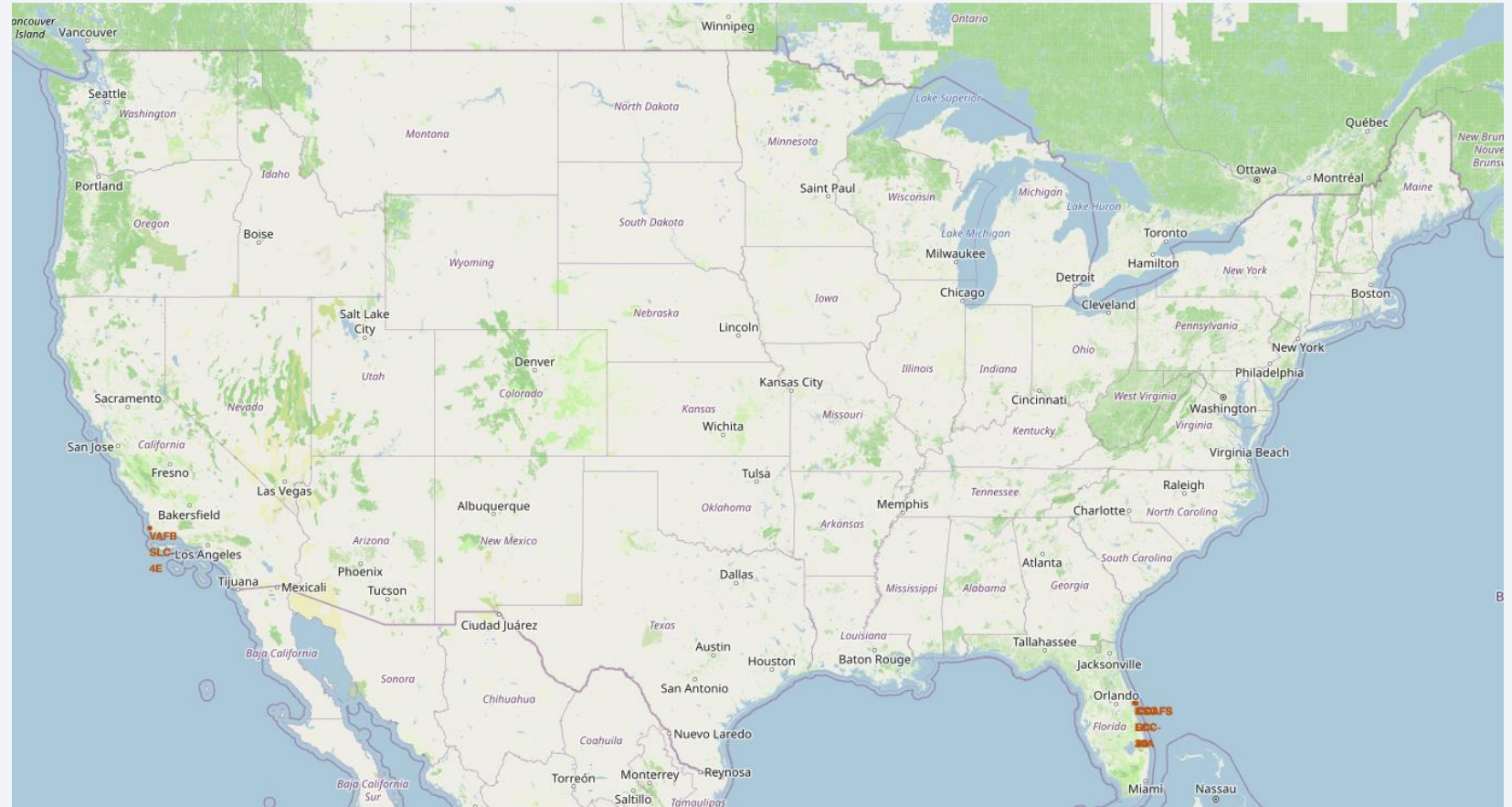
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

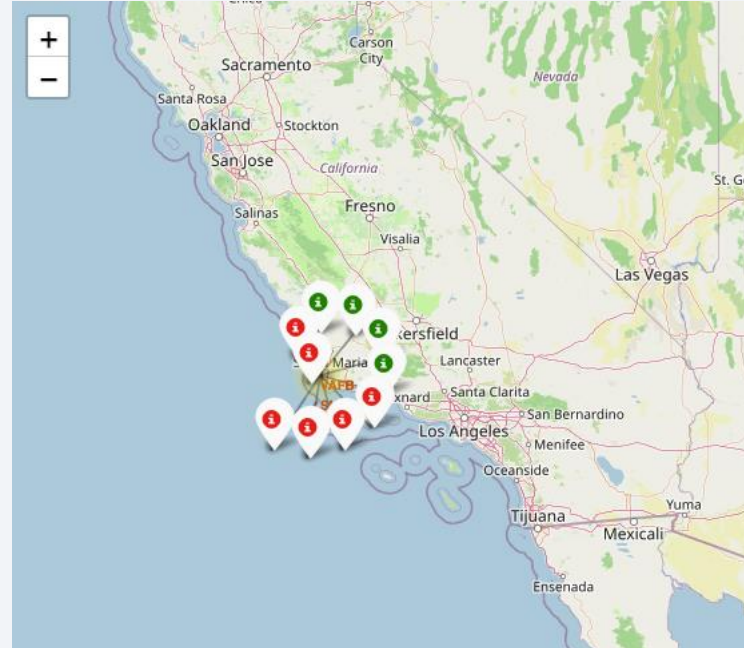
Launch Sites Proximities Analysis

SpaceX Launch Site Locations

- **Explanation:** All 4 launch pads sit near the coast at low latitude (Florida and California), close to the equator (as physically possible from the USA) and open ocean to allow eastward/polar launch corridors clear of populated areas.



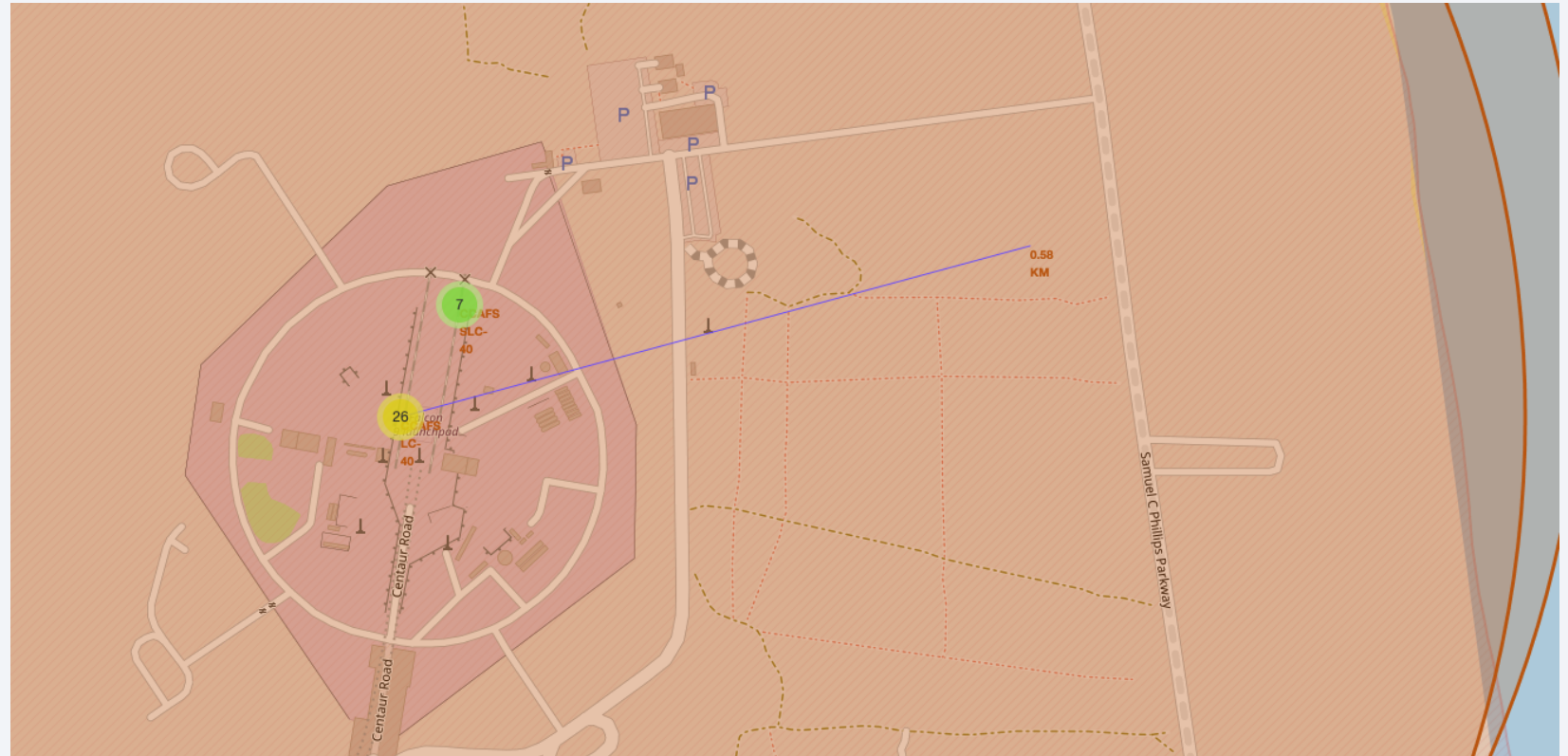
Launch Outcomes by Site (Success vs. Failure)

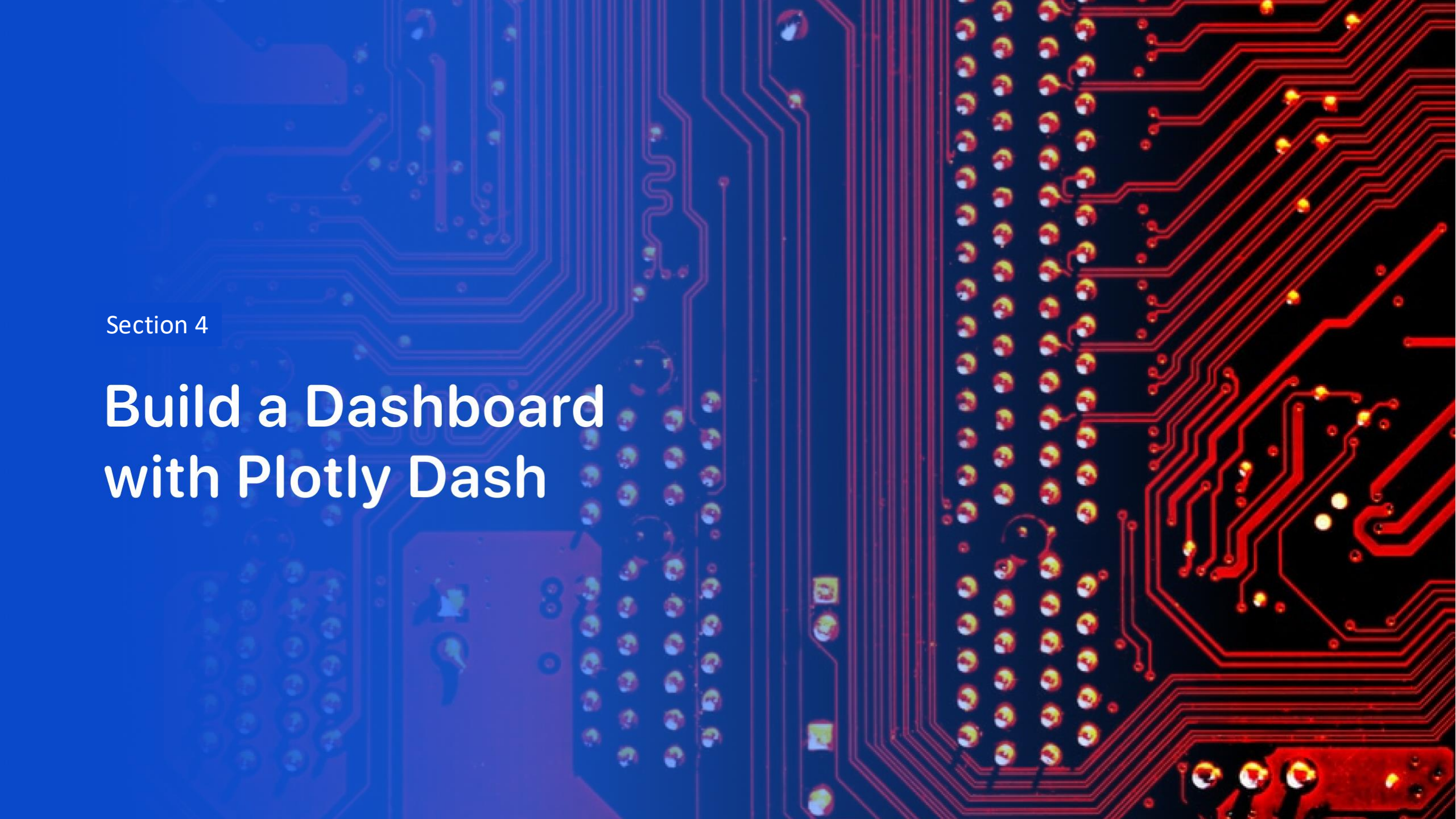


- **Explanation:** KSC LC-39A shows the highest concentration of green (success) markers, while CCAFS LC-40 has a more even mix of success/failure - consistent with its higher launch volume across all booster generations.

<Folium Map Screenshot 3>

- **Explanation:**
CCAFS LC-40 sits just 0.58 km from the coastline and 0.84 km from a highway, but 19.15 km from the nearest city - close enough to infrastructure for logistics, far enough from population centers for safety.

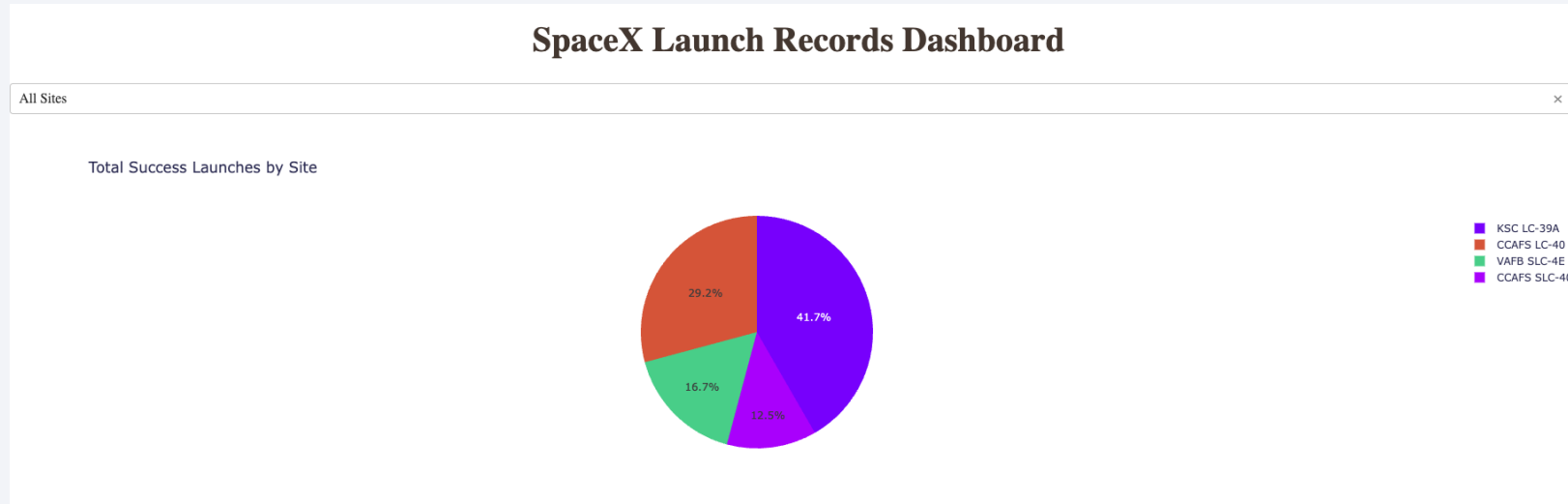




Section 4

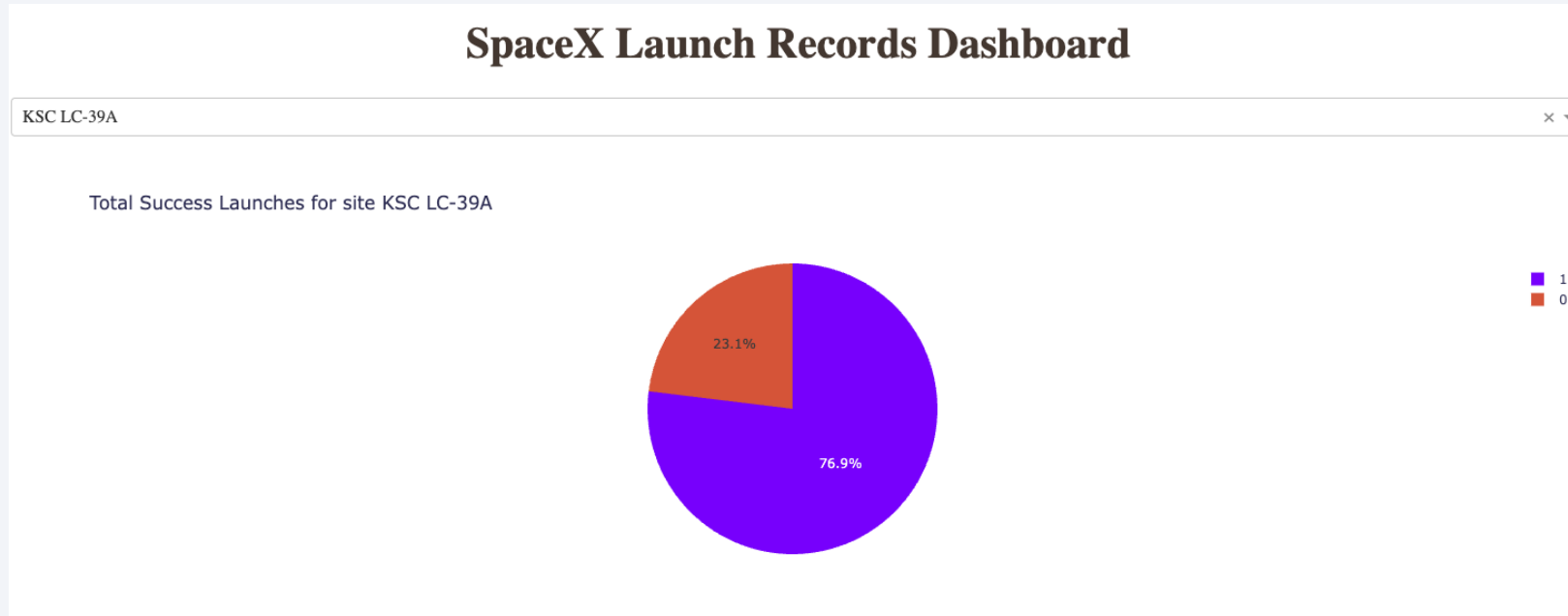
Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>



- **Explanation:** KSC LC-39A contributes 41.7% of all successful launches, followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%), and CCAFS SLC-40 (12.5%).

<Dashboard Screenshot 2>



- **Explanation:** KSC LC-39A has the highest per-site success ratio at 76.9%, versus 42.9% (CCAFS SLC-40), 40.0% (VAFB SLC-4E), and 26.9% (CCAFS LC-40).

Payload vs. Launch Outcome (All Payload Ranges)



- **Explanation:** FT and B4 booster versions succeed across the widest payload range; very low (<1,000 kg) and very high (>9,000 kg) payloads succeed more consistently than the 4,000-6,000 kg mid-range, which shows more mixed outcomes.

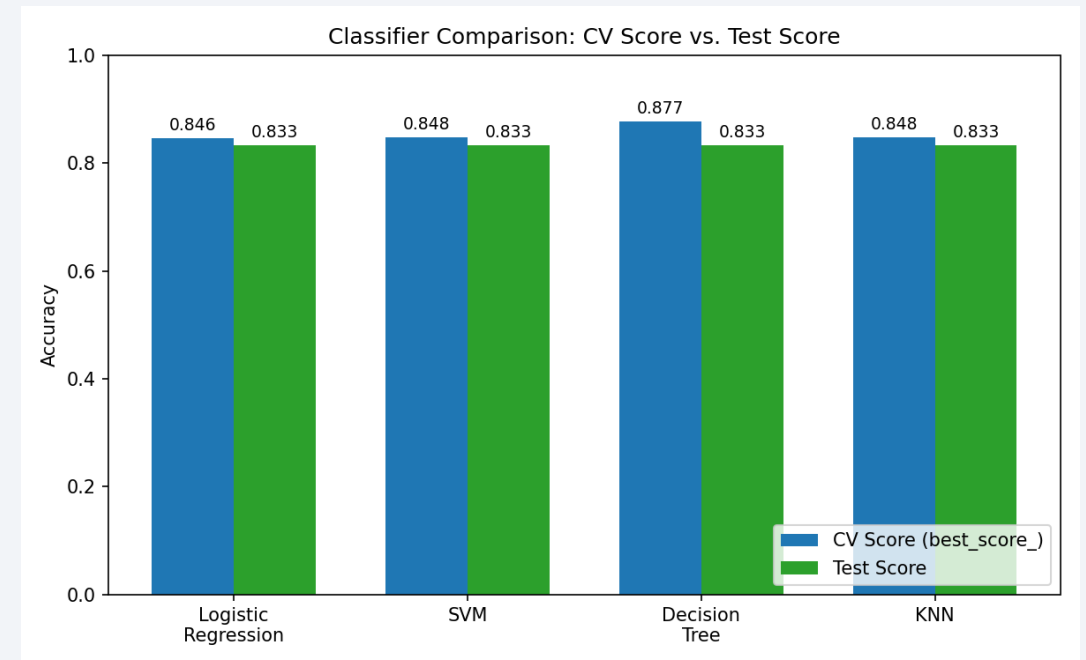


Section 5

Predictive Analysis (Classification)

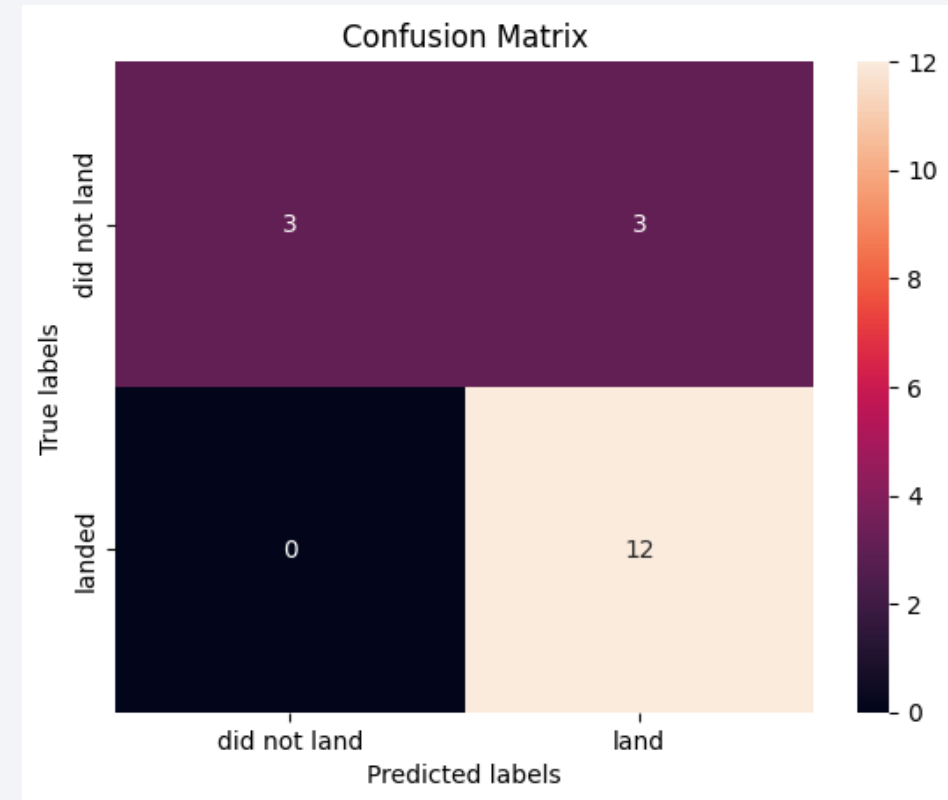
Classification Accuracy

- **Result:** Decision Tree scored highest in 10-fold cross-validation (87.7%), ahead of SVM and KNN (84.8% each) and Logistic Regression (84.6%).
- On the held-out test set, all four models - Logistic Regression, SVM, Decision Tree, and KNN - scored identically at 83.3%. Cross-validation ranking and test performance agree here, with no sign of overfitting.



Confusion Matrix

- **Recommended model:** Decision Tree - ties with Logistic Regression, SVM, and KNN for test accuracy (83.3%) and also leads on cross-validation (87.7%), making it the strongest overall pick.
- **Confusion matrix:** 3/6 correct on non-landings, 12/12 correct on landings - identical across all four models. Note the test set is only 18 flights, so a couple of misclassifications swing accuracy by several points either way.
- **Consistency:** Unlike earlier tuning runs, all four classifiers now generalize equally well from cross-validation to the test set, so CV score is a reliable guide to picking the Decision Tree here.



Conclusions

- Falcon 9 first-stage landing success climbed from 0% (2010-2013) to over 84% (2019-2020) as SpaceX matured its recovery technology.
- KSC LC-39A has the highest per-site success rate (76.9%); GTO is the hardest orbit to land from (~52% success) while ES-L1/GEO/HEO/SSO show 100% success in this dataset.
- Heavier payloads land more reliably in LEO, ISS, and Polar orbits than in GTO, where outcomes are mixed regardless of payload mass.
- Cross-validation and held-out test accuracy agree: the Decision Tree leads on 10-fold CV (87.7%) and ties with Logistic Regression, SVM, and KNN at 83.3% test accuracy - all four models are viable, with the Decision Tree the strongest single recommendation given its CV edge.
- These patterns support the project's premise: launch outcome is predictable from flight/payload characteristics, which could help a competitor estimate SpaceX's launch costs.

Appendix

- **Notebooks (GitHub):** <https://github.com/jjfloreszw/ibmds>
- **Data collection:** 01-introduction/API-data-collection.ipynb, webscraping-data-collection.ipynb, spacex-data-wrangling.ipynb
- **EDA:** 02-EDA/EDA-with-data-viz.ipynb, EDA-with-SQL.ipynb
- **Interactive analytics:** 03-interactive-visual-analytics-dashboard/interactive-visual-analytics-with-folium.ipynb, spacex-dash-app.py
- **Predictive analysis:** 04-predictive-analysis/machine-learning-prediction.ipynb
- **Key datasets:** spacex_web_scraped.csv, dataset_part_3.csv, my_data1.db, spacex_launch_geo.csv, spacex_launch_dash.csv

Thank you!

